

S.No. : 570

NBS 4201

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Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 49905

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No.

1	2	3	4	5	6	7	8	9	10
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B. Tech. Examination, 2024-25

(Even Semester)

DIFFERENTIAL EQUATIONS AND FOURIER ANALYSIS

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION - A

1. Attempt all parts of the following : $8 \times 1 = 8$

(a) Solve $(D^3 + 2D^2 - 3D)y = e^t$, where $D = \frac{d}{dt}$.

(b) Solve simultaneous differential equations :

$$\frac{dx}{dt} = 3y, \frac{dy}{dt} = 3x$$

(c) Show that :

$$P_n(-x) = (-1)^n P_n(x)$$

[P. T. O.

(d) Prove that :

$$J_{-n}(x) = (-1)^n J_n(x)$$

(e) Find the fourier constant a_1 of:

$$f(x) = \pi x, -\pi \leq x \leq \pi$$

(f) Find the fourier constant a_0 for the function :

$$f(x) = \begin{cases} 0, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$$

(g) Solve the partial differential equation :

$$\frac{\partial z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} = 0$$

(h) How many arbitrary constants in the solution of Laplace equation?

SECTION - B

2. Attempt any two parts of the following : $2 \times 6 = 12$

(a) Using method of variation of parameter solve :

$$\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = e^{2x} + x^2$$

(b) Show that :

$$\int_{-1}^1 P_m(x) P_n(x) dx = 0, m \neq n$$

(c) Find half range cosine series for the function $f(x) = (x-1)^2$ in the interval $(0, 1)$. Hence prove that :

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$$

(d) Solve by the method of separation of variables :

$$4 \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = 3u, u = 3e^{-x} - e^{-5x}$$

when $t = 0$.

SECTION - C

Note :- Attempt all questions. Attempt any two parts from each questions. $8 \times 5 = 40$

3. (a) Solve the simultaneous differential equation :

$$\frac{dx}{dt} + 5x - 2y = t, \frac{dy}{dt} + 2x + y = 0$$

being given $x = 0$ and $y = 0$ when $t = 0$.

(b) Solve by changing the independent variable :

$$\frac{d^2 y}{dx^2} + (3 \sin x - \cot x) \frac{dy}{dx} + 2y \sin^2 x = e^{-\cos x} \sin^2 x$$

(c) Solve :

$$(D^2 - 3D + 2)y = x^2 + 2x + 1$$

[P. T. O.]

4. (a) Solve in series the differential equation :

$$2x(1-x) \frac{d^2y}{dx^2} + (5-7x) \frac{dy}{dx} - 3y = 0$$

- (b) Prove that :

$$\frac{1-z^2}{(1-2xz+z^2)^{3/2}} = \sum_{n=0}^{\infty} (2n+1) P_n(x) z^n$$

- (c) Express $J_3(x)$ in terms of $J_0(x)$ and $J_1(x)$.

5. (a) Find the fourier series of :

$$f(x) = \begin{cases} x, & 0 < x < 1 \\ 1-x, & 1 < x < 2 \end{cases}$$

- (b) Expand $f(x) = 2x - 1$ as a sine series in $0 < x < \pi$.

- (c) Find the fourier series of $f(x) = x^2$ in $-\pi \leq x \leq \pi$.

6. (a) A rod of length ℓ with insulated sides is initially at a uniform temperature u_0 . Its ends are suddenly cooled to 0°C and are kept at that temperature. Calculate the temperature function $u(x, t)$.

- (b) Determine the solution of one dimensional wave equation :

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

under the conditions $u(0, t) = 0$, $u(\ell, t) = 0$

for all t , $u(x, 0) = f(x)$ and $\left(\frac{\partial u}{\partial t}\right)_{t=0} = 0$, $0 < x < \ell$.

- (c) Solve :

$$(D+1)(D+D'-1)z = \sin(2x+3y)$$
